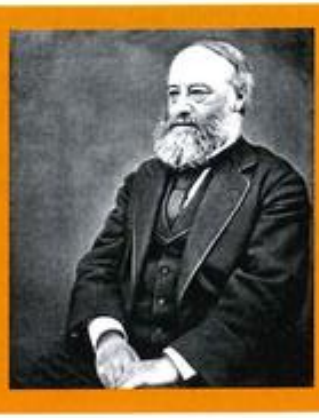
16 WORK AND POWER

Work is calculated by multiplying the force applied by the distance through which the force moves – the bigger the force, or the longer the distance through which it moves, the more work is done. Work always involves an energy transfer. Power is the rate at which energy is transferred, and efficiency is a measure of how much of the input energy to a system is converted to useful output energy. In this chapter you will learn how to calculate the work done in a system and its power as energy is transferred.



▶ Figure 16.1 James Joule (1818–1889) was the son of a wealthy Manchester brewer. He was tutored by James Dalton and carried out scientific research in his own laboratory, built in the basement of his father's home.

LEARNING OBJECTIVES

- Know and use the relationship between work done, force and distance moved in the direction of the force:

  work done = force × distance moved

  $W = F \times d$
- Know that work done is equal to energy transferred
- Know and use the relationship between gravitational potential energy, mass, gravitational field strength and height:

  gravitational potential energy = mass × gravitational field strength × height

  $GPE = m \times g \times h$
- Know and use the relationship:

  kinetic energy = $\frac{1}{2}$ × mass × speed squared

  $KE = \frac{1}{2} \times m \times v^2$
- Understand how conservation of energy produces a link between gravitational potential energy, kinetic energy and work
- Describe power as the rate of transfer of energy or the rate of doing work
- Use the relationship between power, work done (energy transferred) and time taken:

  $$\text{power} = \frac{\text{work done}}{\text{time taken}}$$

  $$P = \frac{W}{t}$$

The unit of energy is named after James Joule. It was Joule who realised that heat was a store of energy. He showed that kinetic energy could be transferred to heat. At that time heat was measured in **calories**.

ENERGY AND WORK

Energy is the ability to do work.

This statement tells us what energy does rather than what energy is. We know that energy is found in a wide variety of different stores but we are really interested in what energy can do – the answer is that energy does work.

We need to define work in a way that is measurable. Some types of work are not easy to calculate the value of. Mechanical work, like lifting heavy objects, is easy to measure: if you lift a heavier object, you do more work; if you lift an

object through a greater distance, again, you do more work. The definition of work in physics is:

work done, W (joules) = force, F (newtons) × distance moved, d (metres)

$$W = F \times d$$

If the force is measured in newtons and the distance through which the force is applied is measured in metres then the work done will be in joules.

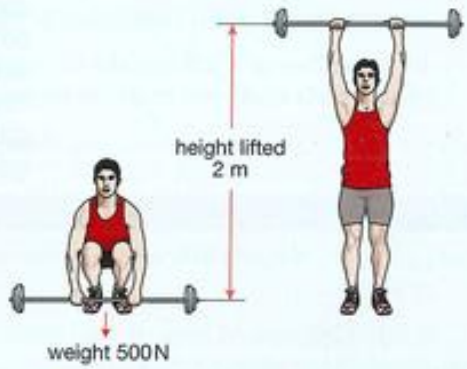
Work done is equal to the amount of energy transferred.

1 J of work done is transferred when a force of 1 N is applied through a distance of 1 m in the direction of the force.

EXAMPLE 1

Figure 16.2 shows a weightlifter raising an object that weighs 500 N through a distance of 2 m. To calculate the work done we use:

$W = F \times d$
$= 500\ \text{N} \times 2\ \text{m}$
$= 1000\ \text{J}$



▲ Figure 16.2 Doing work by lifting a weight

This work done on the weight has increased its energy. This is explained in the section on gravitational potential energy (page 152).

EXAMPLE 2



▲ Figure 16.3 A car travelling at a constant speed doing work

In the example shown in Figure 16.3 the force acting on the car is not accelerating it – instead, it is being used to balance the forces opposing its movement. The resultant force on the car is zero, so it keeps moving in a straight line at constant speed. To work out the work done on the car in one second we substitute the force required, 400 N, and the distance through which it acts in one second, 30 m, in the equation:

$W = F \times d$
$= 400\ \text{N} \times 30\ \text{m}$
$= 12\,000\ \text{J or } 12\ \text{kJ}$

GRAVITATIONAL POTENTIAL ENERGY (GPE)

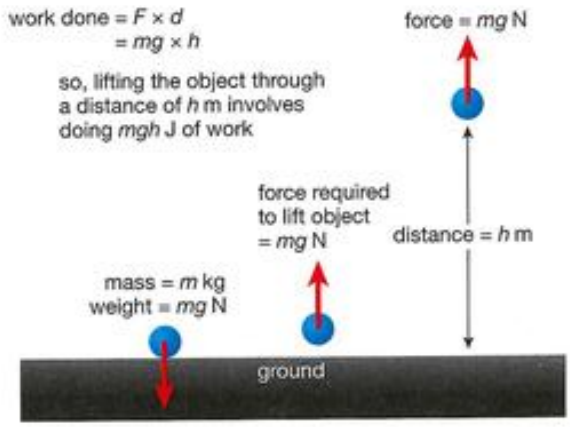
The gravitational potential energy of an object that has been raised to a height, *h*, above the ground is given by:

gravitational potential energy, *GPE* (joules) = mass of object, *m* (kilograms) × gravitational field strength, *g* (newtons per kilogram) × height, *h* (metres)

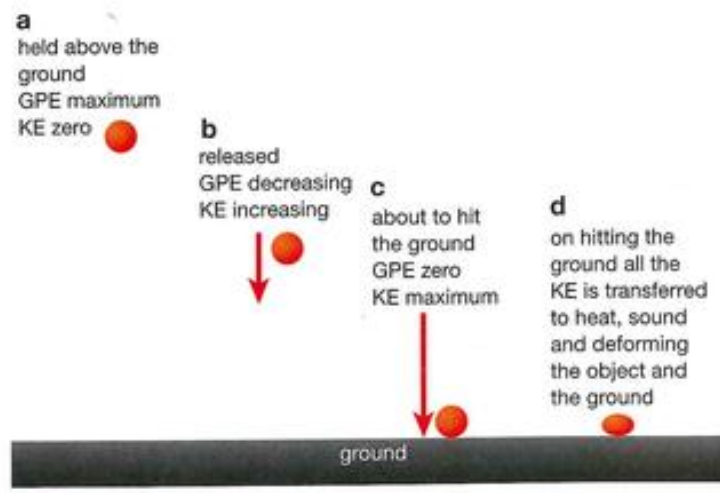
$$GPE = m \times g \times h$$

The change in the GPE of an object will be an increase if we apply a force on it in the opposite direction to the pull of gravity – that is, if we lift it off the ground. When an object falls it loses GPE. To keep things simple, we usually assume that an object has no GPE before we do work on it.

In the weight-lifting example given on page 151, the weightlifter has used some chemical energy to do the work. We know that energy is conserved so what has happened to the chemical energy that the weightlifter used? Some has been transferred to heat in the weightlifter's body. The remainder has been transferred to the weight because he has increased its height in the gravitational field of the Earth. The energy that the weight has gained is called gravitational potential energy or GPE.



▲ Figure 16.4 The work done to lift an object is equal to the GPE the object has at its new height.



▲ Figure 16.5 When a raised object falls, its gravitational potential energy is transferred first to kinetic energy and then to heat and sound.

KEY POINT

Kinetic energy (KE) is the energy stored by moving objects.

In Figure 16.5, we can see the GPE stored by the weight is being transferred to other stores as the weight falls. The weight accelerates because of the force of gravity acting on it, so it gains kinetic energy. When it reaches the ground all the initial GPE is transferred to kinetic energy. When it hits the ground all the movement energy is then transferred to other stores, mainly heat and sound.

REMINDER

Gravitational field strength is the force acting per kilogram on a mass in a gravitational field. The gravitational field strength, g, on the surface of the Earth is approximately 10 N/kg. Since the weight of an object is mg, increase in GPE is a special version of the equation $W = F \times d$, with $F = mg$ and $d = h$.

KINETIC ENERGY, KE

The kinetic energy of a moving object is calculated using the equation:

kinetic energy, KE (joules) = mass, m (kilograms) × speed squared, v^2 (metres squared per seconds squared)

$$KE = \frac{1}{2}mv^2$$

EXTENSION WORK

In Chapter 1 we saw the equation of uniformly accelerated motion:

$v^2 = u^2 + 2as$

This equation allows us to work out the final velocity, v, of an object that has a uniform acceleration, a, and that has accelerated through a distance, s, starting with an initial velocity of u.

If we drop an object from rest its initial velocity, $u = 0$, its acceleration is g m/s^2 and the distance that it falls is usually represented by h, so the equation becomes

$v^2 = 2gh$

Multiplying both sides of the equation by m, the mass of the object, gives:

$mv^2 = 2mgh$ which can be rearranged as $\frac{1}{2}mv^2 = mgh$

showing that the gain in KE of the object is equal to its loss of GPE.

We see that the amount of kinetic energy possessed by a moving object depends on its speed and its mass. As the Earth travels through space, orbiting the Sun, it runs the risk of colliding with chunks of matter that are drawn into the gravitational field of the Solar System. In fact, this is very common. If you have ever seen a shooting star – or, to give it its proper name, a meteor – you have seen the line of light produced as a small piece of space debris (waste) burns up on entering our atmosphere. This is an example of kinetic energy being transferred to heat and light by the friction produced between the air and the object passing through it.



▲ Figure 16.6 Meteors burn up on entering our atmosphere – we see them as 'shooting stars'.

EXTENSION WORK

The meteorite that caused the Arizona crater is thought to have hit the Earth travelling at 11 000 m/s and to have had a mass of 109 kilograms. It hit the ground with an energy equivalent to a 15 megaton hydrogen bomb, 1000 times greater than the atomic bomb dropped on Hiroshima at the end of the Second World War.



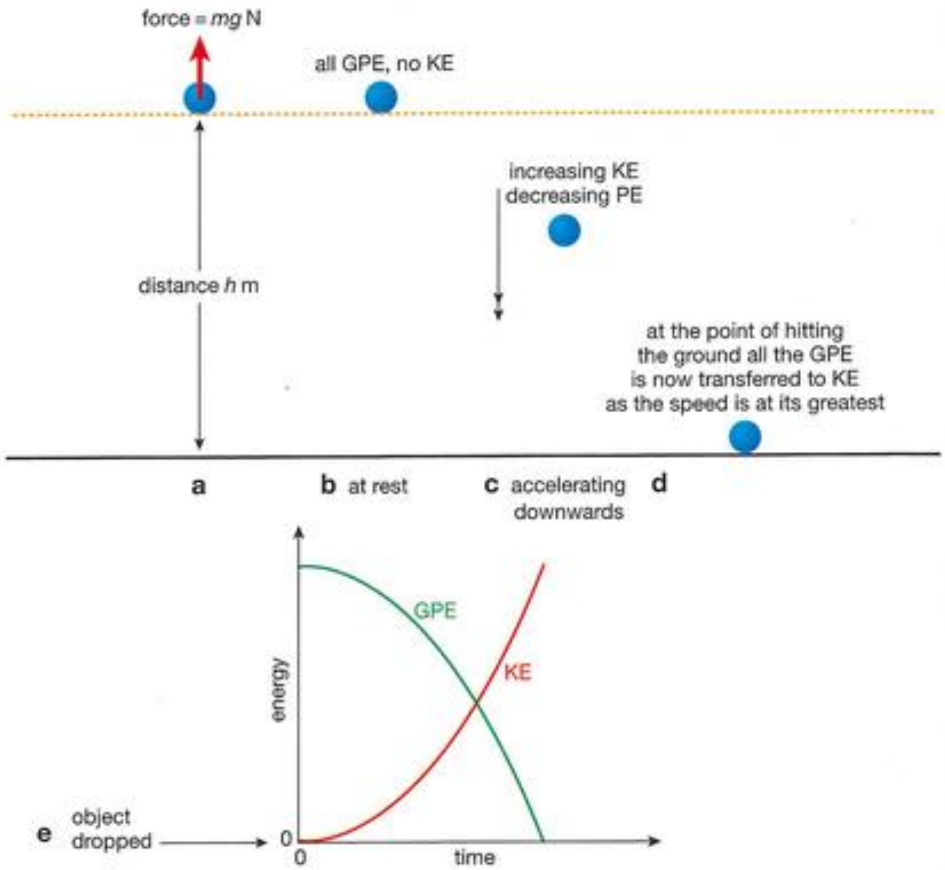
▲ Figure 16.7 This crater was created when a meteorite collided with Earth in Arizona.

EXAMPLE 3

Calculate the kinetic energy carried by a meteorite of mass 500 kg (less than that of an average-sized car) hitting the Earth at a speed of 1000 m/s.

$$KE = \frac{1}{2}mv^2$$
$$= \frac{1}{2} \times 500\ \text{kg} \times (1000\ \text{m/s})^2$$
$$= 250\,000\,000\ \text{J (or 250 MJ)}$$

CALCULATIONS USING WORK, GPE AND KE



▲ Figure 16.8 GPE and KE of a falling object: **a** doing work to lift an object; **b** all GPE; **c** GPE transferring to KE during fall; **d** all KE at end of fall; **e** graph showing relationship between GPE and KE as the object falls

Work transfers energy to an object: $W = Fd$. An object of mass, m, weighs ($m \times g$) newtons so the force, F, needed to lift it is mg (Figure 16.8). If we raise the object through a distance h, the work done on the object is $mg \times h$. This is also the gain in GPE.

When the object is released, it falls – it loses GPE, but gains speed and so gains KE. At the end of the fall, all the initial GPE of the stationary (not moving) object has been transferred to the KE of the moving object. The graphs in Figure 16.8 show how the GPE of the object is changing into KE as it falls. The sum of the two graphs is always the same. Energy is conserved, so the loss of GPE is equal to the gain in KE.

$$\text{work done lifting object} = \text{gain in GPE} = \frac{\text{gain in KE of the object just}}{\text{before hitting the ground}}$$

EXAMPLE 4



▲ Figure 16.9 A rollercoaster ride

In a rollercoaster ride the truck falls through a height of 17 m. Calculate the truck's speed at the bottom of this fall. (Take g = 10 N/kg)

If we assume that all the GPE of the truck at the top of the ride is transferred to KE at the bottom we can use the equation:

$$GPE = KE$$

$$mgh = \tfrac{1}{2}mv^2$$

The mass, m, of the truck appears on both sides of the equation, so it cancels out:

$$gh = \tfrac{1}{2}v^2$$

Substituting h = 17 m and g = 10 N/kg:

$$17\text{ m} \times 10\text{ N/kg} = \tfrac{1}{2}v^2$$

$$v^2 = 2 \times 17\text{ m} \times 10\text{ N/kg}$$

$$v = \sqrt{(2 \times 17\text{ m} \times 10\text{ N/kg})}$$

$$= 18.44\text{ m/s (about 66 kph)}$$

Some of the GPE the truck had at the start will be transferred to heat and sound, so the speed at the bottom of the fall will be a little slower than this.





▲ Figure 16.10 a James Watt (1736–1819) was a Scottish engineer who improved the performance of the steam engine and can be said to have started the Industrial Revolution – the beginning of the machine age. b A model of a steam engine that transfers the heat energy of steam to movement. Watt's engines were used to pump water out of mines.

POWER

Power is the rate of transfer of energy or the rate of doing work.

James Watt (Figure 16.10a) is remembered as the inventor of the steam engine and is said to have been inspired by watching the lid on a kettle being forced up by the pressure of the steam forming inside. Neither story is accurate, but what is true is that Watt, working in partnership with Matthew Boulton, developed improvements to the steam engine that made it a commercial product and completely changed industry and transport.

The SI unit of power is named in honour of James Watt. The watt (W) is the rate of transfer or conversion of energy of one joule per second (1 J/s).

$$\text{power, } P\text{ (watts)} = \frac{\text{work done, } W\text{ (joules)}}{\text{time taken, } t\text{ (seconds)}}$$

$$P = \frac{W}{t}$$

ACTIVITY 1

▼ PRACTICAL: INVESTIGATE YOUR POWER OUTPUT

You may have done a simple experiment involving running upstairs to measure your output power. You do work as you raise your GPE, and to find your power output in watts you divide the work done by the time taken. The experiment is shown in Figure 16.11. Notice that calculating the work you do against gravity using force × distance works just as well as using the equation for *GPE* (mass × gravitational field strength × height).

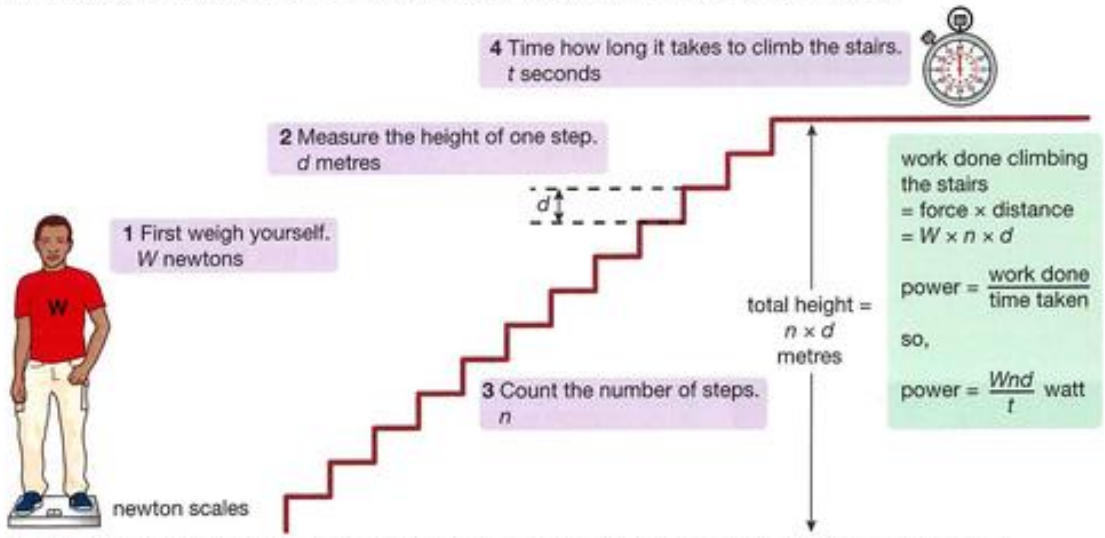


Figure 16.11 An experiment to measure your output power

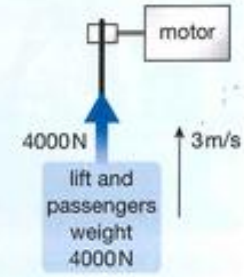
If you don't have scales measuring in newtons, simply multiply your mass in kg by 10 to convert to newtons.

! Wear suitable footwear and only allow one person at a time on the staircase. Ensure the stairs are dry, in good condition and free of any obstacles.

A more convenient way of raising your GPE and getting to a higher floor in a building is to take a lift. The lift will transfer its energy input, usually electrical, to kinetic energy and then, if you are going up, to GPE. As usual, unwanted energy transfers are inevitable – sound and heat will be produced. If we know the weight of the lift and its contents and the height through which it moves, we can calculate the work done in the usual way. If we measure the time that the lift journey takes we can then calculate the power output of the lift motor. (Strictly this will be the useful power output – it will not take account of the wasted power due to unwanted energy transfers.)

EXAMPLE 5

If a lift and passengers have a combined weight of 4000 N and the lift moves upwards with an average speed of 3 m/s then what is the useful power output of the lift motor?



▲ Figure 16.12

To keep the lift moving upwards at a steady speed, the lift motor must provide an upward force to balance the weight of the lift. This is 4000 N. In each second, this force is applied through a vertical distance of 3 m, so:

work done per second = 4000 N × 3 m/s

= 12 000 J/s

= 12 000 W

CHAPTER QUESTIONS

More questions on work, power and efficiency can be found at the end of Unit 4 on page 168.

In the questions below, where necessary, take the strength of the Earth's gravity to be 10 N/kg.

SKILLS CRITICAL THINKING (5)

1 James Joule showed that heat is a store of energy. He did this by showing that heat can be produced by using mechanical energy.

a Give an example of a process in which kinetic energy is transferred to heat.

b Describe how heat energy can be transferred to either kinetic energy or gravitational potential energy.

(4) 2 a State the SI unit of work.

(5) b Define the unit of work.

SKILLS PROBLEM SOLVING (7) c How much work is done in each of the following situations?

i A bag of six apples each weighing 1 N is lifted through 80 cm.

ii A rocket with a thrust of 100 kN travels to a height of 200 m.

iii A weightlifter raises a mass of 60 kg through a height of 2.8 m.

iv A lift of mass 200 kg lifts three people of mass 50 kg each through a distance of 45 m.

(8) 3 Water from a hydroelectric power station reservoir is taken from a reservoir (artificial lake) at a height of 800 m above sea level to turbines (engines) in the power station itself. The power station is at sea level. The reservoir holds 200 million (2×10^8) litres of water. If a litre of water has a mass of 1 kg, how much gravitational potential energy is stored in the water in the reservoir?

SKILLS CRITICAL THINKING 4 a State how to calculate the kinetic energy stored by a moving object.

SKILLS PROBLEM SOLVING b Work out the kinetic energy of the following:

i a man of mass 80 kg running at 9 m/s

ii an air rifle pellet of mass 0.2 g travelling at 50 m/s

iii a ball of mass 60 g travelling at 24 m/s.

(9) 5 A catapult fires a stone of mass 0.04 kg vertically upwards. If the stone has an initial kinetic energy of 48 J, how high will it travel before it starts to fall back to the ground?

6 If a coin is dropped from a height of 80 m, how fast will it be travelling when it hits the ground? State any assumptions you may need to make.

SKILLS CRITICAL THINKING (5) 7 Define power and state its unit.

SKILLS PROBLEM SOLVING 8 A person with a mass of 40 kg runs upstairs in 12 s. The stairs have 20 steps and the height of each step is 20 cm.

(6) a How much does the person weigh, in newtons?

(5) b What is the total height that the person has climbed?

(7) c Calculate how much work is done in climbing the stairs.

d What is the power output of the person running up the stairs?

9 A drag car, of mass 500 kg, accelerates from rest to a speed of 144 km/h in 5 s.

(6) a What is its final speed in:

i m/h (metres per hour)

ii m/s?

(8) b What is the increase in KE of the drag car?

c What is the average power developed by the drag car's engine?